

F517 — the substrate decomposition $n_C = \text{rank} + N_c = \text{'two crossings + three commitments'}$, surfaced by Casey's 3-ball question and grounded in the banked Peirce decomposition (T2511). The five substrate directions of D_IV^5 split, via the Jordan-algebra Peirce decomposition of the rank-2 spin factor, into: (a) the rank = 2 diagonal idempotents $[V_{11} \oplus V_{22}, \text{each 1-dim}] = \text{the TWO BOUNDARY SHEETS} = \text{Casey's 'two medium changes'}$ (enter the boundary / leave the boundary); and (b) the transverse Peirce space V_{12} , of dimension $a = n_C - \text{rank} = 3 = N_c$ (the short-root multiplicity, T2511/F368; $SO(V_{12}) = SO(3)$) = the THREE COLOR COMMITMENTS. So $n_C = 5 = \text{rank}(2) + N_c(3) = \text{two crossings + three commitments} — \text{NOT a numerical coincidence but the Peirce structure of the substrate Jordan algebra. This is what the cone-tip 3-ball (F516) IS: at the } \nu=0 \text{ vertex (the } \Gamma_ \Omega \text{ pole / cone tip), the phase circle } S^1 \text{ (the } SO(2) \text{ weight direction), having no idempotent to anchor it, opens into the transverse } V_{12} = \text{the } a=N_c=3 \text{ commitment directions } \rightarrow \text{ a 3-ball, whose volume coefficient } 4/3 \text{ normalizes the vertex residue } \Gamma(-a/2)=\Gamma(-3/2)=4\sqrt{\pi/3} \text{ down to } \sqrt{\pi} \text{ (F516; the '3' of the 3-ball = } a = N_c = \text{ the short-root multiplicity, over-determined as } a = n_C - \text{rank} = N_c \text{). PHYSICAL READING (Casey's): the substrate's 5 dimensions are 2 boundary-crossings (absorb/emit — the rank, the two Shilov sheets) + 3 committed color channels (the transverse Peirce space); the tau (sitting exactly AT the } \nu=0 \text{ vertex) is the first observable whose mass-integral opens into the full 3-commitment ball, which is why its cone-tip term carries the 3-ball volume. HONEST FRICTION (to reconcile, flagged by$

Keeper): T2511 currently reads V_{12} as ‘relabeling, not growth’ — so the F516 ‘ S^1 blows up into the 3-ball’ must be reconciled with the Peirce structure: is it a genuine blow-up (the $SO(2)$ circle opening into the $SO(3)$ -transverse V_{12} at the degenerate vertex) or a relabeling of already-present directions? The PARAMETER decomposition ($n_C = \text{rank} + N_c$, Peirce-grounded) is SOLID/banked; the explicit vertex blow-up ($S^1 \rightarrow V_{12}$ 3-ball) is the open geometric step (same open piece as F516’s cone-tip). NET: a clean, banked structural result — the substrate splits as two boundary-crossings + three color-commitments, $n_C = \text{rank} + N_c$ — independent of whether the tau integral fully closes; it names WHY the cone-tip is a 3-ball (it’s the transverse Peirce/commitment space) and ties Casey’s ‘two medium changes’ ($\text{rank}=2$) and ‘three commitments’ ($N_c=3$) to one Peirce decomposition. For Keeper ($n_C = \text{rank} + N_c = \text{two crossings} + \text{three commitments}$) is the Jordan-algebra Peirce decomposition of the rank-2 spin factor (T2511 banked: $V_{11} \oplus V_{22}$ diagonal = rank = 2 sheets; V_{12} transverse = a = $N_c = 3$, $SO(3)$); this is your Section 4a; the parameter split is solid, the explicit $S^1 \rightarrow V_{12}$ vertex blow-up is the open geometric step (reconcile with T2511’s ‘relabeling not growth’ reading); the cone-tip 3-ball of F516 IS V_{12}), Casey (your 3-ball question surfaced the clean split: the substrate’s five directions = two boundary crossings (the rank, your ‘two medium changes’ enter+leave) + three color commitments (the transverse Peirce space $V_{12} = N_c = 3$); the tau sits at the vertex where the phase circle opens into those three commitment directions, which is why its mass carries a 3-ball; five = two crossings + three commitments, and it’s the Peirce structure, not a coincidence),

Elie (the transverse $V_{12} = \text{SO}(3)$ short-root space (T2511/F368) is the 3-ball's home; for the tau build, the $S^1 \rightarrow V_{12}$ blow-up at the vertex is the piece to construct — is it growth or relabeling per T2511?), Grace ($n_C = \text{rank} + N_c = 2 + 3$ as a banked Peirce identity for the scorecard's structural spine), Cal.

Lyra (Claude Opus 4.8)

2026-07-12 Sunday (date-verified)

F517 — the substrate is two crossings plus three commitments: $n_C = \text{rank} + N_c$

Casey's question about the cone-tip 3-ball — “is it the three commitments? what are the three components?” — surfaced a clean structural decomposition of the substrate, and it's grounded in a banked theorem (T2511, the Peirce decomposition).

The decomposition

The five substrate directions of D_{IV}^5 split, via the Jordan-algebra Peirce decomposition of the rank-2 spin factor, into:

$$n_C = 5 = \underbrace{\text{rank} = 2}_{V_{11} \oplus V_{22}, \text{ diagonal idempotents}} + \underbrace{a = N_c = 3}_{V_{12}, \text{ transverse Peirce space}}.$$

- The rank = 2 are the two diagonal idempotents ($V_{11} \oplus V_{22}$, each 1-dimensional) — the two boundary sheets, Casey's “two medium changes” (enter the boundary / leave the boundary).
- The $a = n_C - \text{rank} = 3 = N_c$ are the transverse Peirce space V_{12} (short-root multiplicity, T2511/F368; $\text{SO}(V_{12}) = \text{SO}(3)$) — the three color commitments.

So five = two crossings + three commitments — not a numerical coincidence, but the Peirce structure of the substrate Jordan algebra.

What the cone-tip 3-ball is

This names the F516 result. At the $\nu=0$ vertex (the cone tip, the Γ_Ω pole), the phase circle S^1 (the $\text{SO}(2)$ weight direction) — having no idempotent to anchor it — opens into the transverse V_{12} = the $a = N_c = 3$ commitment directions $\rightarrow$ a 3-ball. Its volume coefficient $4/3$ normalizes the vertex residue $\Gamma(-a/2) = \Gamma(-3/2) = 4\sqrt{\pi}/3$ down to $\sqrt{\pi}$. The “3” of the 3-ball is $a = N_c$ = the short-root multiplicity (over-determined: $a = n_C - \text{rank} = N_c$). The tau, sitting exactly *at* the vertex, is the first observable whose mass-integral opens into the full 3-commitment ball — which is why its cone-tip term carries the 3-ball volume.

Honest friction (to reconcile)

T2511 currently reads V_{12} as “relabeling, not growth.” So the F516 picture — “the S^1 blows up into the 3-ball” — must be reconciled with the Peirce structure: is it a genuine blow-up (the $\text{SO}(2)$ circle opening into the $\text{SO}(3)$ -transverse V_{12} at the degenerate vertex), or a relabeling of directions already present? The parameter decompo-

sition ($n_C = \text{rank} + N_c$, Peirce-grounded) is solid/banked; the explicit vertex blow-up ($S^1 \rightarrow V_{12}$ 3-ball) is the open geometric step — the same open piece as F516’s cone-tip.

Plain-language version

You asked whether the little “3-ball” that showed up in the tau’s mass calculation is really “the three commitments,” and what the three are. The answer is yes, and it’s cleaner than I expected. Our space has five directions, and there’s a standard mathematical way (the Peirce decomposition) to split them — and it splits exactly as $2 + 3$: two of the directions are the “boundary-crossing” directions (the *enter* and *leave* you pointed at weeks ago), and the other three are the color directions — the three things the substrate “commits.” So *five = two crossings plus three commitments*, and it’s forced by the algebra, not a coincidence. The tau lives right at the sharp tip of the space, and at that tip the leftover “clock” direction opens up into those three commitment directions — a little 3-dimensional ball — which is exactly the volume factor ($4/3$) that made the tau’s mass come out right. The one thing I still owe: showing rigorously that the clock direction *opens into* those three (a real growth) rather than just getting relabeled — our own earlier theorem is cautious about that, and it’s the honest next step. But the split itself — two crossings, three commitments — is solid.

— Lyra, 2026-07-12. F517: $n_C = \text{rank} + N_c = \text{two crossings} + \text{three commitments}$ (Peirce decomposition, T2511 banked). D_{IV}^5 rank-2 spin factor: diagonal $V_{11} \oplus V_{22} = \text{rank} = 2$ (the two boundary sheets / Casey’s two medium changes) + transverse $V_{12} = a = n_C - \text{rank} = 3 = N_c$ ($SO(3)$, the three color commitments). The F516 cone-tip 3-ball IS V_{12} : at $\nu=0$ the S^1 opens into the 3 transverse commitment directions, vol $4/3$ normalizes $\Gamma(-3/2)=4\sqrt{\pi/3} \rightarrow \sqrt{\pi}$. Parameter split solid/banked; explicit $S^1 \rightarrow V_{12}$ blow-up = open step (reconcile with T2511 ‘relabeling not growth’). Keeper Section 4a.